Xylem & Phloem

Exam code: 0610 & 0970

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The xylem & phloem

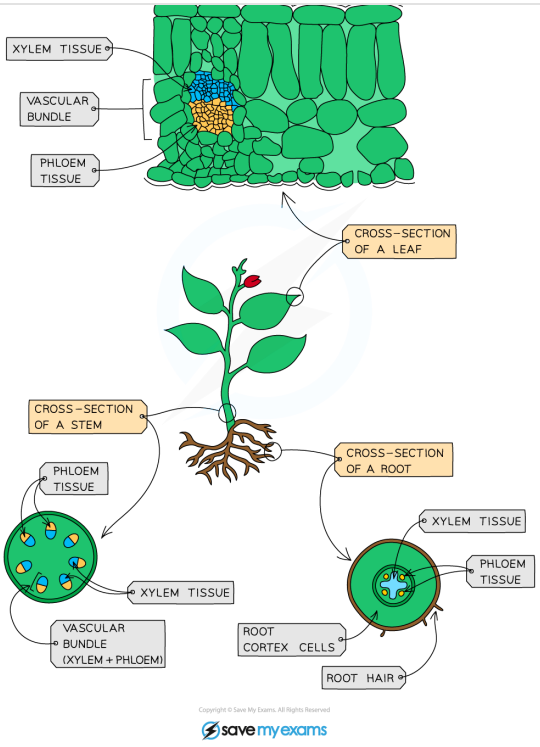
What is the function of the xylem and phloem

- Plants contain two types of transport vessel:
  - **Xylem vessels** – transport water and minerals (pronounced: zi-lem) from the roots to the stem and leaves
  - **Phloem vessels** – transport food materials (mainly sucrose and amino acids) made by the plant from photosynthesising leaves to non-photosynthesising regions in the roots and stem (pronounced: flow-em)
- These vessels are arranged throughout the root, stem and leaves in groups called **vascular bundles**

Xylem and phloem diagram

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



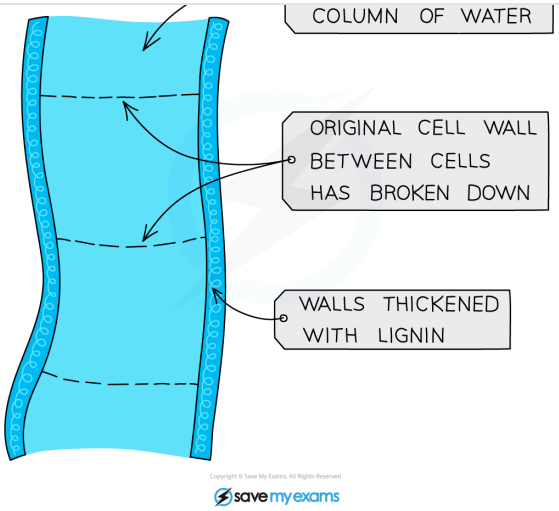
Vascular tissue in a dicotyledonous plant

Examiner Tips and Tricks

If you are asked to identify the xylem or phloem in a diagram showing a cross-section of a root, stem or leaf just remember that **xylem is always on the inside and phloem is always on the outside.**

Adaptations of xylem vessels: Extended

- Function: transport tissue for water and dissolved mineral ions
- Adaptations:
  - Cells joined end to end with **no cross walls** to form a long continuous tube
  - Cells are essentially **dead,** without cell contents, to allow **free passage of water**
  - Outer walls are **thickened** with a substance called **lignin,** strengthening the tubes, which helps **support** the plant



Xylem cells lose their top and bottom walls to form a continuous tube through which water moves through from the roots to the leaves

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